Kim Hyoyeol

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Professional Summary

Engineer transitioning from semiconductor process engineering to cloud/ML systems. Built and validated ML prototypes and web APIs (PyTorch, FastAPI) and conducted performance-oriented analysis (latency/profiling mindset, bottleneck identification) in project settings. Leverage a strong troubleshooting and data-driven root-cause analysis background from high-stakes manufacturing environments, and communicate clearly in Korean and English.

Professional Experience

- **BEOL Process Integration Engineer** 2024
  Samsung Electronics, Hwaseong · Sub-5nm Logic Process
  - Supported BEOL process integration for sub-5nm logic; analyzed yield data and performance metrics using statistical methods to identify optimization pathways
  - Developed understanding of on-chip SRAM architecture, memory bandwidth constraints, and hardware-software codesign tradeoffs in specialized compute systems
  - Conducted systematic root-cause analysis on yield-limiting factors; applied data-driven methodology to characterize process windows and design margins
- **Etch Process Development Engineer** Jan 2022 – 2024
  SEMES (Semiconductor Equipment Co.) · Far-BEOL Etch Process
  - Developed and qualified plasma etch processes on 300mm production equipment; achieved target uniformity and selectivity specifications through systematic optimization
  - Performed equipment characterization; analyzed performance bottlenecks and applied statistical methods to resolve instability issues
  - Collaborated on process requirement specification with foundry customer teams

Education and Current Engagement

- **B.S. Advanced Materials Science & Engineering** **Sungkyunkwan Univ. (SKKU)**
  GPA: 3.65/4.5 — Capstone: Copper Nanoparticle Paste Improvement for Automotive Semiconductor Die-attach Process *2022*
- **Microsoft AI School — Advanced ML Engineering Program** 2025 – Present
  - Advanced ML engineering program focused on production ML systems, model evaluation, and cloud deployment on Azure

Certifications

- **Microsoft Certified: Azure AI Fundamentals (AI-900)**

Technical Skills

- **ML Systems & Inference**: inference profiling mindset (latency/throughput, bottleneck analysis), cost-aware optimization concepts, evaluation/monitoring approach in projects
- **AI & Deep Learning**: PyTorch (custom architectures, CNNs, model optimization, hyperparameter tuning), inference optimization (batching, scheduling, latency profiling), model evaluation (Bayesian methods, cross-validation, production monitoring)
- **Cloud Infrastructure & Tools**: Azure ML, Azure AI Foundry (Studio), Azure OpenAI, Azure AI Search, Blob Storage; Docker, Git
- **Software Development**: Backend (FastAPI, REST API design, basic JWT auth patterns), Frontend (React, TypeScript), project-level system design and deployment
- **Hardware-Informed Optimization**: on-chip SRAM architecture and memory bandwidth optimization, hardware-software codesign tradeoffs and constraint-aware design, statistical root-cause analysis and data-driven optimization, performance bottleneck identification and resolution
- **Programming Languages**: Python, TypeScript
- **Languages**: Korean (Native), English (Advanced)

Selected Project Highlights

Condensed technical summary for AI Platform / Inference Engineering roles

Project A: SNAP-Q — AI Vehicle Damage Assessment

2-week sprint: multi-model vision pipeline producing repair cost estimates with uncertainty (LinkedIn post)

- **Role**: led Mask R-CNN damage segmentation; prioritized accuracy on real-world camera images.
- **Training/infra**: CUDA-based PyTorch training in **Azure ML Studio** on scalable GPU infrastructure.
- **Inference integration**: built **Bayesian fusion logic** combining Mask R-CNN segmentation with Custom Vision classification.
- **Outcome**: end-to-end pipeline producing repair cost predictions **with confidence intervals**.

Project B: HONEYPOT — AI Knowledge Handover Chatbot & Documentation System (HR SaaS)

RAG-based platform to convert implicit individual knowledge into durable organizational assets across transfers/leaves/turnover (LinkedIn post)

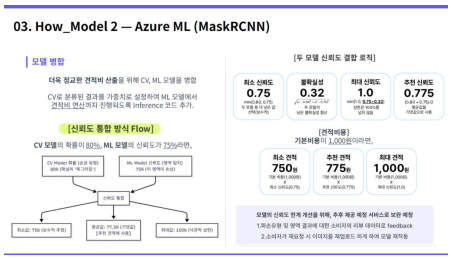
- **Problem**: knowledge gaps recur across not only departures but also transfers, role changes, and extended absences.
- **My role (PM/Architecture)**: designed modular service architecture for diverse HR scenarios; defined a handover **JSON schema** configurable by department/role.
- **RAG strategy**: engineered preprocessing pipeline (context-preserving summaries, scenario-specific metadata/tags, robust chunking, HR-workflow fields like people/sections) to improve retrieval quality beyond naive indexing.
- **B2B infra**: leveraged Azure services for enterprise-grade security/scalability; maintained self-hosting-ready code and templates.

Project C: Perceptron-Gated Noise Simulation for 6T SRAM Reliability

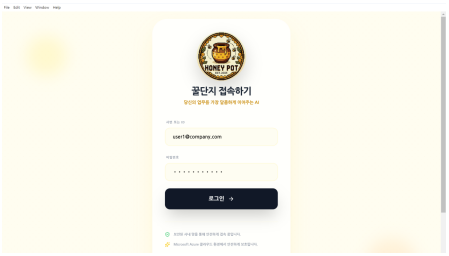
Hybrid physics-ML framework to model transistor-level noise under PVT conditions with scalable Monte Carlo analysis (LinkedIn post)

- **Motivation**: TCAD-based Monte Carlo can take **10+ hours**, slowing reliability iteration.
- **Core idea**: learn **(Temperature, Voltage) → noise weights** and inject perceptron-weighted noise into a 6T SRAM loop.
- **Implemented**: SNM/butterfly curves (FF/TT/SS), Pelgrom variability + confidence intervals, Euler–Maruyama SDE, retention (quasi-potential), NBTI/HCI + Weibull degradation.
- **Engineering**: Streamlit real-time tuning; GPU-compatible scaling goal; physics-constrained losses (SNM/BER thresholds).

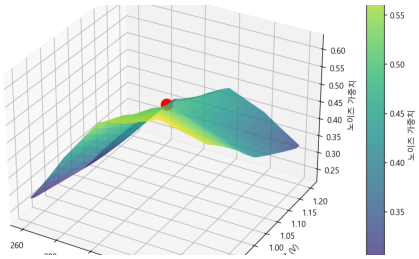
Project Visuals



SNAP-Q fusion flow



HONEYPOT (login/UI)



SRAM noise surface

Tools & Capabilities

Languages	Python
AI / ML	PyTorch, Mask R-CNN, Bayesian uncertainty & fusion, RAG (metadata-driven retrieval)
Systems	Data preprocessing pipelines, async inference orchestration, caching/batching concepts, monitoring/evaluation mindset
Cloud	Azure (AI services / ML Studio), secure B2B-oriented deployment considerations
Modeling	Monte Carlo, SDE (Euler–Maruyama), reliability/aging models (NBTI/HCI)